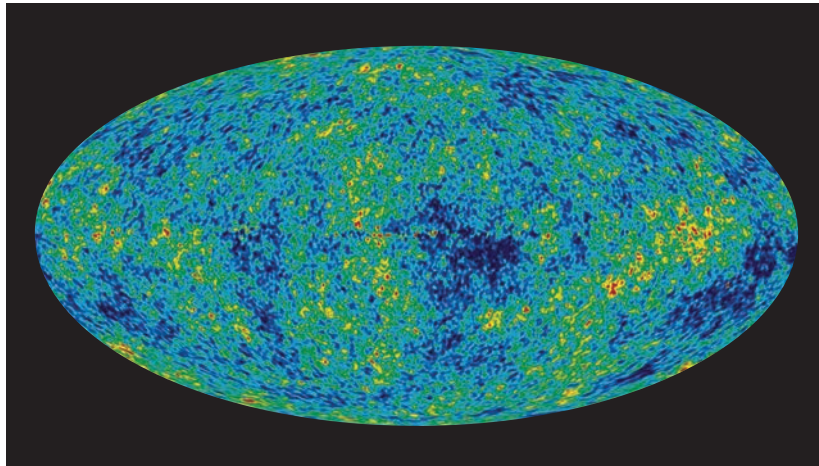


7. In 1802, spectra from stars and galaxies were first observed.

In the 1940s the Steady State Theory of the origin of the Universe was proposed by Sir Fred Hoyle and other scientists.

In 1964, American astronomers discovered there was electromagnetic interference coming from all parts of the sky at all times. This is shown in the image below.



- (a) State the name of the region of the electromagnetic spectrum that is recorded on the image shown above. [1]

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- (b) Explain the evidence that led to an alternative, different theory of the origin of the Universe to the Steady State Theory. [6 QER]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1										4 He Helium 2																																																																															
7 Li Lithium 3					9 Be Beryllium 4					11 B Boron 5					12 C Carbon 6					14 N Nitrogen 7					16 O Oxygen 8					19 F Fluorine 9					20 Ne Neon 10																																																						
23 Na Sodium 11					24 Mg Magnesium 12															27 Al Aluminium 13					28 Si Silicon 14					31 P Phosphorus 15					32 S Sulfur 16					35.5 Cl Chlorine 17					40 Ar Argon 18																																												
39 K Potassium 19					40 Ca Calcium 20					45 Sc Scandium 21					48 Ti Titanium 22					51 V Vanadium 23					52 Cr Chromium 24					55 Mn Manganese 25					56 Fe Iron 26					59 Co Cobalt 27					59 Ni Nickel 28					63.5 Cu Copper 29					65 Zn Zinc 30					70 Ga Gallium 31					73 Ge Germanium 32					75 As Arsenic 33					79 Se Selenium 34					80 Br Bromine 35					84 Kr Krypton 36				
86 Rb Rubidium 37					88 Sr Strontium 38					89 Y Yttrium 39					91 Zr Zirconium 40					93 Nb Niobium 41					96 Mo Molybdenum 42					99 Tc Technetium 43					101 Ru Ruthenium 44					103 Rh Rhodium 45					106 Pd Palladium 46					108 Ag Silver 47					112 Cd Cadmium 48					115 In Indium 49					119 Sn Tin 50					122 Sb Antimony 51					127 I Iodine 53					128 Te Tellurium 52					131 Xe Xenon 54				
133 Cs Caesium 55					137 Ba Barium 56					139 La Lanthanum 57					179 Hf Hafnium 72					181 Ta Tantalum 73					184 W Tungsten 74					186 Re Rhenium 75					190 Os Osmium 76					192 Ir Iridium 77					195 Pt Platinum 78					197 Au Gold 79					201 Hg Mercury 80					204 Tl Thallium 81					207 Pb Lead 82					209 Bi Bismuth 83					210 Po Polonium 84					210 At Astatine 85					222 Rn Radon 86				
223 Fr Francium 87					226 Ra Radium 88					227 Ac Actinium 89																																																																															

Key

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number